Networking Practice Set

TCP congestion control · Delays and switching · Subnetting · True/false. The answer key with workings starts on page 3.

> **Tips:** For TCP, draw a table: burst | sent | window afterwards | threshold. For subnetting, use the 5 steps: cutting octet → block = 256 − octet → start → subnet (then 0s) → broadcast (next start − 1, then 255s).

Part A: TCP congestion control

A1. MSS = 2 KB, threshold = 16 KB, receiver's advertised window = 24 KB. What is the congestion window after the **8th** transmission?

A2. Slow start on a line with a **10 ms** RTT and no congestion. Receiver's window = 36 KB, threshold = 20 KB, MSS = 4 KB. How long until the first full window can be sent?

A3. The congestion window is **46 KB** when a timeout occurs. MSS = 4 KB. What is the congestion window after **5** successful transmissions?

A4. MSS = 3 KB, threshold = 30 KB, receiver's window = 45 KB. There is a timeout on the **9th** transmission. What is the congestion window after the **14th** transmission?

Part B: Delays and switching

B1. A —L1— B —L2— C. L1 runs at **100 Mb/s** and L2 at **400 Mb/s**. A sends P1 (**2,000 B**), then P2 immediately after all of P1 is on L1. What is the smallest size of P2 (in bytes) so that there is no queueing at B?

B2. Send a **0.5 MB** message across **4 intermediate nodes**. Circuit setup = 500 ms, propagation = 5 ms per hop, packet size = 10,000 B, data rate = 8 Mb/s at each node. Ignore headers and queueing. How many times bigger is the circuit switching delay than the packet switching delay? (Give 2 decimal places.)

Part C: Subnetting

C1. A network has the mask **255.255.248.0**. What is the maximum number of hosts?

C2. IP **172.16.45.9**, mask **255.255.240.0**. Find the subnet address and the broadcast address.

C3. IP **135.20.66.130**, mask **255.255.255.192**. Find:
a. the class
b. the network ID
c. usable hosts per subnet
d. the number of subnets
e. the broadcast address of this subnet

C4. With mask **255.255.255.224**, are **192.168.1.40** and **192.168.1.60** in the same subnet?

C5. IP **88.44.130.77**, mask **255.255.192.0**. What is the broadcast address?

C6. Which of these are **valid host** addresses?
a. 200.1.1.127 with mask 255.255.255.128
b. 10.5.0.0 with mask 255.255.0.0
c. 150.10.12.0 with mask 255.255.248.0
d. 220.300.1.1 with mask 255.255.255.0

Part D: True or false?

D1. An IP address with **/27** will have a broadcast address ending in 31, 63, 95, 127, 159, 191, 223 or 255. **T / F**

D2. A Class B network has 65,536 usable hosts. **T / F**

D3. Removing the Network PCI from a Network PDU gives the Transport PDU. **T / F**

D4. In TCP, after a timeout the threshold is set to half of the receiver's advertised window. **T / F**

D5. 10.10.10.255 is a valid host with mask 255.255.254.0. **T / F**

D6. ARP maps a domain name to a MAC address. **T / F**

Answer key

Try every question before you look.

Part A

A1. 24 KB

Sent: 2, 4, 8, 16 (reached the threshold), then +2: 18, 20, 22, 24. After the 8th (sent 24), the window stays at **24**, because that's the receiver's window and it can't go higher.

A2. 70 ms

Sent: 4, 8, 16, **20** (doubling 16 would give 32, capped at 20), then +4: 24, 28, 32, **36** (full). That's 7 bursts before the full one, so 7 × 10 = **70 ms**.

A3. 31 KB

New threshold = 46 ÷ 2 = 23, and the window resets to 4. Sent: 4, 8, 16, **23** (capped), 27. After the 5th success the window is 27 + 4 = **31**.

A4. 27 KB

Sent: 3, 6, 12, 24, **30** (capped), 33, 36, 39, then the 9th sends 42 and times out. Threshold = 21, window = 3. Then: 10th sends 3, 11th sends 6, 12th sends 12, after which the window is **21** (24 would pass the threshold, so capped). 13th sends 21 → 24. 14th sends 24 → **27**.

Part B

B1. 500 B

Condition: time to send P2 on L1 ≥ time to send P1 on L2. So $S_2 \div 100 \geq 2000 \div 400$, giving $S_2 \geq 2000 \times 100 \div 400$ = **500 B**. Propagation cancels out.

B2. ≈ 1.81

Hops = 4 + 1 = 5, so propagation = 5 × 5 = 25 ms.

Circuit: 500 (setup) + $0.5\times10^6\times8 \div 8\times10^6$ = 500 ms (transmission) + 25 = **1,025 ms**.

Packet: $T_{pkt} = 10{,}000 \times 8 \div 8\times10^6$ = 10 ms. N = 500,000 ÷ 10,000 = 50 packets. Delay = (50 + 4) × 10 + 25 = **565 ms**.

Ratio = 1,025 ÷ 565 = **1.81**.

Part C

C1. 2,046 hosts

248 = `11111000` (5 ones), so 8 + 8 + 5 = /21. Host bits = 11, so $2^{11} - 2$ = **2,046**.

C2. Subnet 172.16.32.0 · Broadcast 172.16.47.255

The cut is in the 3rd octet (240), so block = 16. 45 ÷ 16 = 2.8, which becomes 2, and 2 × 16 = **32**. Subnet = 172.16.32.0. Next start = 48, so 48 − 1 = 47, giving broadcast 172.16.47.255.

C3.

(a) Class	(b) Network ID	(c) Hosts	(d) Subnets	(e) Broadcast
B	**135.20.0.0**	**62**	**1,024**	**135.20.66.191**

135 is in 128–191, so Class B (/16). The mask is /26, so host bits = 6 and $2^6 - 2$ = 62. Subnet bits = 26 − 16 = 10, so 2^{10} = 1,024. Block 64: 130 is in 128–191, so the broadcast is .191.

C4. Yes, same subnet
Block = 256 − 224 = 32, so the parts are 0, 32, 64… Both 40 and 60 fall in **32–63**, so both are in subnet 192.168.1.32.

C5. 88.44.191.255
The cut is in the 3rd octet (192), so block = 64. 130 is in 128–191. The subnet is 88.44.128.0, and the broadcast is 191 then 255 = 88.44.191.255.

C6. Only (c) is valid
(a) /25 means blocks of 128, so .127 is the **broadcast** of 0–127 ✗ · (b) the host part 0.0 is all zeros, so it's the **network** address ✗ · (c) block 8: 12 is in 8–15, the subnet is 150.10.8.0, and 150.10.12.0 isn't the subnet or the broadcast, so it's **valid** ✓ · (d) 300 > 255, so it isn't a real address ✗

Part D

D1. True : /27 gives blocks of 32, and each broadcast is a multiple of 32 minus 1.

D2. False : $2^{16} - 2 =$ **65,534**, because the all-0s and all-1s addresses are reserved.

D3. True : removing the Network PCI leaves the Network SDU, and the Network SDU is the Transport PDU.

D4. False : the threshold is set to half the **current congestion window**.

D5. True : /23 gives block 2 in the 3rd octet, so 10 is in 10–11. The subnet is 10.10.10.0 and the broadcast is 10.10.11.255, so .10.255 is an ordinary host.

D6. False : DNS maps a name to an IP, and ARP maps an IP to a MAC.